

Predicting H1N1 Vaccine Uptake

Machine Learning Insights for Public Health Decision-Making

Goal:

Use survey data to understand the factors influencing vaccination and help public health organizations improve vaccination outreach.

Public Health Problem

During the 2009 H1N1 pandemic(swine flu), vaccination was the most effective tool to control the spread of the virus.

However, many individuals chose not to vaccinate.

Public health organizations need to understand:

- Why some people vaccinate
- Why others hesitate

Machine learning can help identify the strongest predictors of vaccination behavior.

Dataset Overview

Source: National 2009 H1N1 Flu Survey

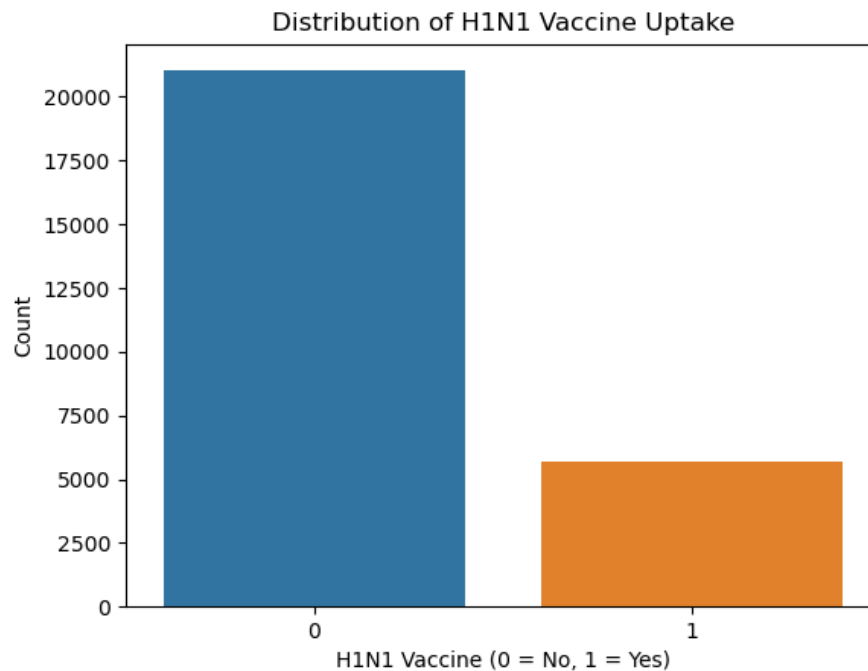
Dataset contains:

- 26,000+ individuals
- Demographic characteristics
- Preventive behaviors
- Vaccine perceptions
- Doctor recommendations

Target variable:

Whether the respondent received the H1N1 vaccine.

Vaccination Distribution



This chart shows the distribution of individuals who received the vaccine.

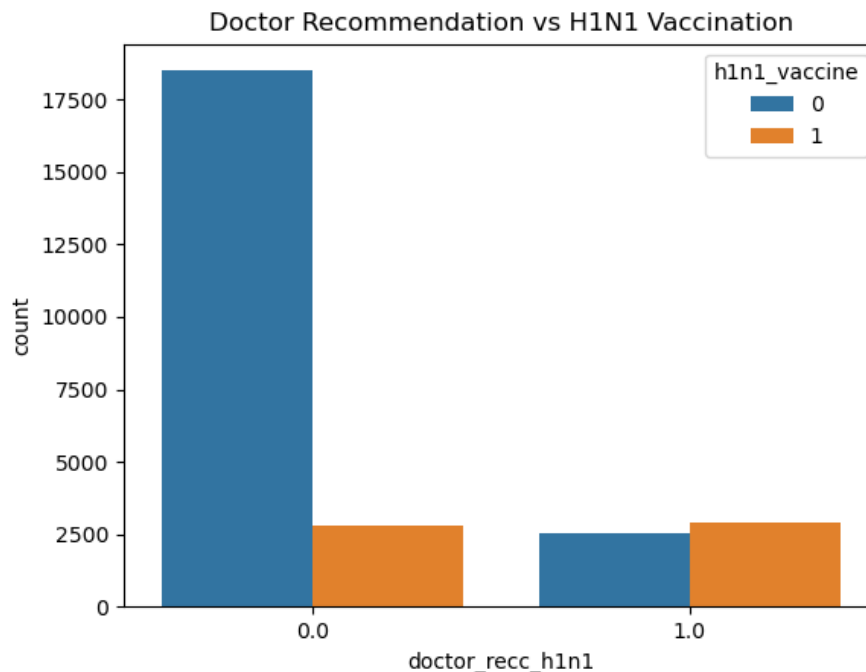
Observation:

Most individuals in the dataset did NOT receive the vaccine.

Implication:

The dataset is imbalanced, meaning the model must carefully distinguish between vaccinated and non-vaccinated individuals rather than simply

Impact of Doctor Recommendation

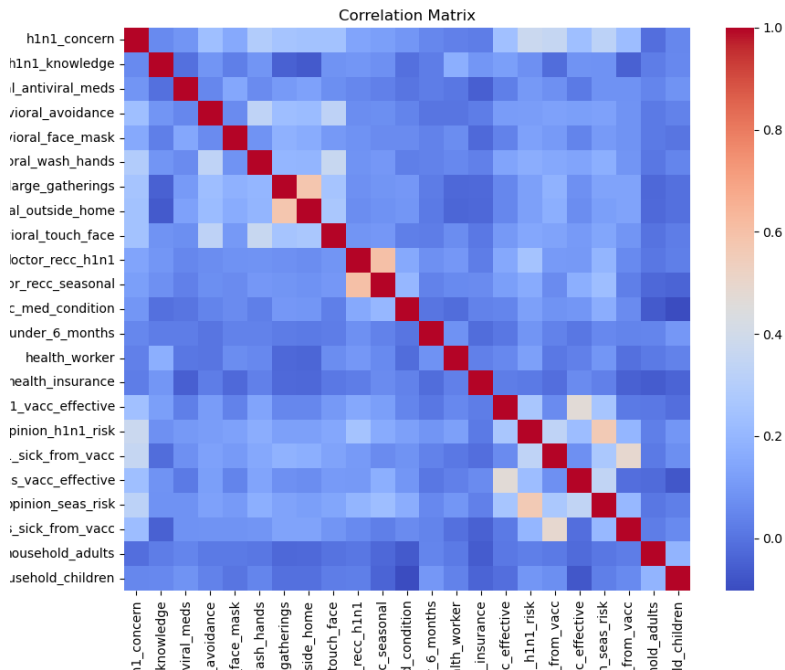


This visualization compares vaccination rates between people who received a doctor recommendation and those who did not.

Observation:

Individuals whose doctors recommended the vaccine were significantly more likely to vaccinate.

Correlation Between Variables



The correlation matrix shows relationships between variables.

Observation:

Most correlations are moderate rather than extremely strong.

Implication:

Vaccination behavior is influenced by multiple factors such as risk perception, vaccine beliefs, and doctor recommendations,

Machine Learning Models

Three classification models were evaluated:

- Logistic Regression
- Decision Tree
- Random Forest

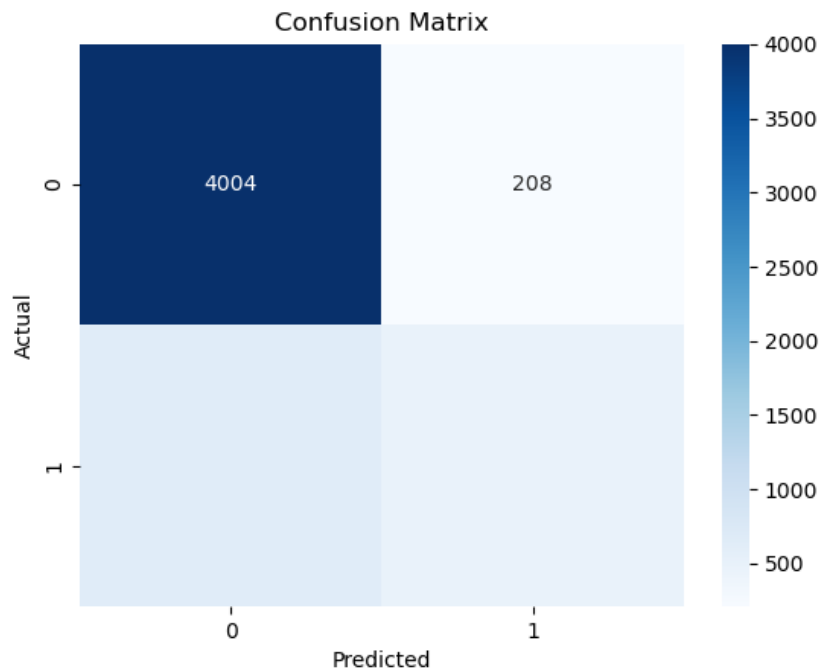
Evaluation metrics included:

Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC.

Goal:

Identify the model that best predicts vaccination behavior while remaining interpretable for public health stakeholders.

Confusion Matrix



The confusion matrix shows how well the model predicts vaccination.

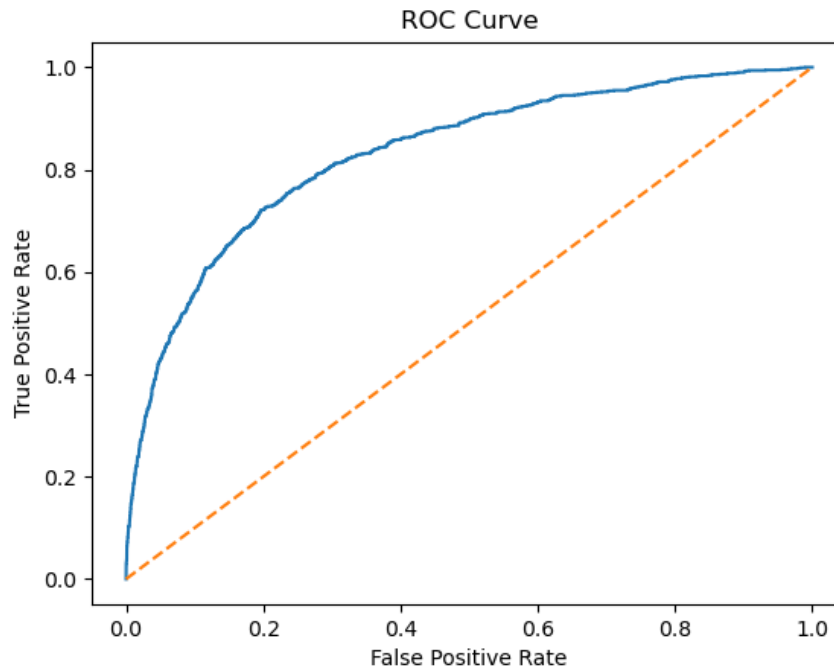
Observation:

The model correctly identifies most individuals who did not receive the vaccine, but it is less accurate in predicting vaccinated individuals.

Implication:

This occurs because fewer individuals in the dataset received

ROC Curve (Model Performance)

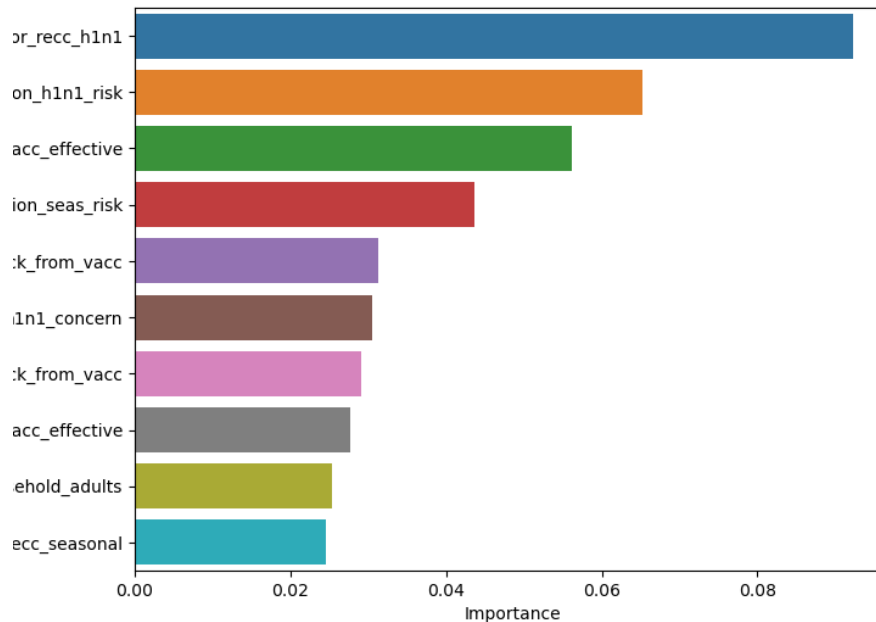


The ROC curve measures how well the model separates vaccinated and non-vaccinated individuals.

Observation:

The curve stays well above the diagonal line, indicating strong predictive performance.

Key Predictors of Vaccination



Feature importance shows which variables most influence predictions.

Top predictors include:

- Doctor recommendation
- Perceived H1N1 risk
- Belief in vaccine effectiveness
- Concern about the virus

Key Insights

1. Doctor recommendation is the strongest driver of vaccination.
2. Individuals who perceive higher risk are more likely to vaccinate.
3. Healthcare workers show higher vaccination rates.
4. Older populations tend to vaccinate more often.

These patterns highlight the importance of trust in healthcare providers and public health messaging.

Recommendations for Public Health

- Encourage healthcare providers to actively recommend vaccines.
- Increase public awareness of influenza risks.
- Develop targeted campaigns for populations with low vaccination rates.
- Expand workplace vaccination programs in healthcare environments.

Conclusion

Machine learning helps identify the key factors influencing vaccination behavior.

Doctor recommendation, risk perception, and vaccine beliefs were among the most important predictors.

These insights can help public health organizations design more effective vaccination campaigns and increase uptake.